Diabetes Risk Prediction System using Machine Learning with Explainable AI (SHAP) and Streamlit Dashboard

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Abstract— Diabetes mellitus is a long-term disease that is highly prevalent in the global society and which has severe health and economic consequences. The timely intervention and better patient outcomes can be achieved through early and precise prediction of the risk of diabetes. In this paper, we will introduce an intelligent system of predicting diabetes based on using machine learning algorithms, such as Logistic Regression, Decision Tree, Random Forest, and XGBoost, which will be trained on a high-scale (100,000 patient records) clinical dataset. The most successful model is automatically chosen according to the accuracy and ROC-AUC score and implemented in the form of an interactive Streamlit web application. SHAP (SHapley Additive exPlanations) is used to make predictions have feature-level explanations to improve their clinical interpretation and trust. A risk report and clinical recommendations is dynamically created and personalized to each patient. The experimental findings indicate that XGBoost can perform better in terms of classification, which is a promising model to use in the initial screening of diabetes and healthcare decision-making.

Keywords — Diabetes Prediction, Machine Learning, XGBoost, Random Forest, SHAP Explainability, Clinical Decision Support, Streamlit, Healthcare AI

I. INTRODUCTION

Diabetes mellitus is a long-term disease that is highly prevalent in the global society and which has severe health and economic consequences. The International Diabetes Federation (IDF) believes that more than 537 million adults are already living with diabetes in the world today and this number is likely to increase exponentially in the next few decades. The most common type of diabetes is type 2 diabetes, which is thought to cause about 90-95 percent of all cases, and is highly related to lifestyle choices: obesity, physical inactivity, and unhealthy eating habits - which, however, can be easily avoided when detected early and with lifestyle changes.

Conventional diagnosis processes are based on laboratory tests (fasting blood glucose, HbA1c, etc.) which are typically performed when the patient already shows symptoms. This reactive method results in late diagnosis and higher chances of complications such as cardiovascular disease, kidney failure, neuropathy and blindness. It is thus urgently required that proactive and data-driven screening tools be developed that will be able to detect high-risk people before they develop clinical symptoms.

The growing volume of Healthcare datasets due to the rise in EHR systems and medical data generated means that new opportunities exist for utilizing the data produced with Machine Learning (ML) technologies (i.e., using EHRs). From the arising opportunities, the identification of patterns in the relationship between clinical features and patient treatment outcomes (based upon their relationship) is possible through ML. ML technology can detect complicated relationships between clinical parameters and the results of patients' treatment better than any other method previously employed in statistics, thus allowing the creation of dependable and scalable forecasts about a patient's response to their treatment than could be done through other traditional statistical techniques.In this work, we suggest a diabetes prediction system that trains and compares four ML classifiers; Logistic Regression, Decision tree, Random Forest and XGBoost, on a 100,000-patient records dataset. The optimal model is automatically chosen according to performance indicators and implemented as an easy-to-use web application. Most importantly, explainability (SHAP-based) is integrated to make clear, feature-based explanations of every prediction, closing the gap between model performance and clinical trust.

II. LITERATURE SURVEY

The use of machine learning techniques in the field of health care has allowed for much better predictions and diagnoses of chronic conditions like diabetes. Over the past few years, multiple studies have been conducted that focused on the creation of predictive models that were not only able to produce accurate predictions, but that also contained an adequate level of interpretability necessary for clinical decision making.

Barman et al. [1] created an interpretable machine learning model to predict diabetes risk utilizing large-scale population data. Their research highlights the importance of transparency in predictive models, specifically the fact that having an understanding of how features contribute to the

predicted outcome is vitally important for clinician acceptance and trust.

Netayawijit et al. [2] presented a framework that combines machine learning with data balancing techniques (e.g., SMOTE) and incorporates SHAP into their overall framework for the purpose of providing interpretability to the predictions. The outcome of this framework is an increased level of classification performance as well as addressing class imbalance and providing insights into which features are most important.

Barzegar et al. [3] developed a machine learning-based early diabetes detection system using a combination of different classifiers along with explainable AI technique implementations. This study demonstrated that predictive accuracy along with interpretability would increase the confidence and reliability of healthcare predictive systems.

In paper [4] authors presented a novel method for diagnosing diabetes through a self-explanatory model which provides both accurate results as well as understandable models for the users to use it and trust it through increased user engagement. The purpose of this study is to show how the use of an explanation mechanism within a system improves user engagement then trust in the model(s).

AutoML was integrated with XAI methods into a framework for providing automated model selection and interpretability for clinical DSS's was proposed by Hasan et al [5]. Their research detailed just how important the integration of automated model selection with interpretability can be advantageous for clinical decision support.

Lundberg & Lee [6] developed the SHAP method (SHapley Additive exPlanations) which is extensively used for providing explainability for ML. The SHAP method will assign a contribution value (positive / negative) to each of the features in the dataset therefore providing users with both global and local explanations of how the ML model made its prediction. In the healthcare domain these aspects are vital as they help to build user confidence and trust in the output of the ML model.

XGBoost, developed by Chen & Guestrin [7], is a scalable and efficient algorithm for gradient boosting that has been validated to outperform traditional methods in various classification problems including predicting diseases. Also due to its scalability and ability to deal with the large number of records in a dataset and to take into account multiple features within the dataset and how these features interact with one another makes it an excellent choice for analyzing healthcare data sets.

There are numerous publicly available datasets from which researchers can use when training and testing their ML models for predicting diabetes such as the diabetes dataset on Kaggle [8]. These databases assist researchers in training and testing their models on real word databases and improve on their ability to generalize and demonstrate robustness for predicting diabetes.

Research Gap—While the current research has helped improve precision of diabetes prediction and the interpretability of the models used, many issues are still unresolved. The majority of current models prioritize either accuracy of prediction or explain ability but do not provide a solution that combines the two in a unified framework. Many models also do not offer user-friendly interfaces that allow people who do not have a technical background to understand the prediction results and use them in making health care decisions. In addition to these issues, most existing solutions do not adequately address real time predictions, interactivity of visualizations or personalized recommendations for clinical decision-making. Therefore, there is a need for a comprehensive solution that provides a reliable, interpretable, user-centered decision-support system for healthcare providers by integrating accurate, machine learning models and explainable AI techniques within an interactive web-based platform.

III. EXISTING SYSTEM

Current techniques for diabetes prediction predominantly utilize conventional clinical diagnostic techniques and rudimentary machine learning algorithms. Standard diagnostic methods include blood analyses performed in a laboratory routine (like fasting blood glucose and / or HbA1c tests) after patients have developed diabetes symptoms. Testing for diabetes should be performed before a person has any signs or symptoms of diabetes (the proactive approach). Consequently, individuals who have the diagnosis of diabetes (delayed diagnosis) develop the complications associated with diabetes.

Many models have been developed to create diabetic risk assessment algorithms with machine learning and classic algorithms (Logistic Regression, Decision Trees, Support Vector Machine, etc). They outperform conventional diagnostic methods in terms of their predictive performance; however, they all have the same limitations. Many of the models were developed purely with the intent to create prediction accuracy and therefore, do not allow health workers or patient's easy access to understand them.

Moreover, few models use Explainable AI (XAI) techniques; thus, they are 'black box' models. These black-box models provide little if any resource as to how to interpret the model output and identify potential risk factors. Another major shortcoming observed in most of the existing ML-based automated diabetes risk prediction systems is that they do not have easy-to-read user interfaces that allow non-technical users to access the system easily. Finally, the absence of real-time predictions and visualisation helps to reduce the effectiveness of these systems in real-world health care applications.

However, despite having advanced ability to predict accurately; these systems fall short of being interpretable, usable and able to provide comprehensive decision support which shows that there is still much development left for better integrated and more advanced solutions. More specifically, with regard to interpretability; users do not trust the decisions made because they do not see how the features of the system were used to arrive at the decision. The system is difficult for the users who are not technically adept to use because it lacks an engaging, user-friendly interface.Most systems also do not offer the user the ability to do short-term data analysis, nor provide the user personalized recommendations based on their individual patient record. Consequently, there is a lack of opportunity for patients to receive adequate preventative care or to be involved in decisions about their own care with this system.

IV. PROPOSED METHODOLOGY

Using a minimal amount of inputted patient information and machine learning algorithms, the proposed system of a web-based application was developed to quickly determine risk for falling in an individual. This application can also provide a rational explanation as to why an individual received their particular score.Four machine learning models were trained on the given dataset and the model that had the most success was kept for future use. The machine learning models included Logistic Regression, Decision Trees, Random Forest and XGBoost algorithms. The machine learning algorithm XGBoost resulted in the best performing model and was ultimately selected for deployment. The web application runs on Streamlit and does not require end users to have technical skills to access or use it.

The prediction pipeline is composed of a series of functional modules, each module performs a discrete function in the processing of input data before the computational analysis is performed by the machine learning model. Each functional module provides a logical progression of functions, the sequence of which produces an individual risk score along with an explanation of how the risk score was derived. Each functional module will be outlined in the following section.

A. Input Module for Users

The users of these modules provide their health information through an input module. Streamlit is being used to provide the interface in a sidebar panel, and users will interact with this interface to provide a value for age, BMI, HbA1c level, blood glucose level, hypertension status, heart disease status, gender, and smoking history via sliders or dropdown menus. The input controls will be clearly labeled with the input range for each entry clearly defined to ensure that users do not enter unrealistic values by mistake. After entering all of their input, the user will click the predict button to send the data to be processed.

B. Preprocessing Input Data

Raw input cannot be directly inputted into the model due to the categorical fields and numerical fields being of different scales. Gender and smoking history categorical fields are transformed into two new categorical fields with binary values through one-hot encoding (the same encoding of the fields that was used during training) prior to being entered into the model. After these categorical fields are encoded, the numerical values for age, BMI, HbA1c, and blood glucose will be rescaled using the StandardScaler object (specifically the instance saved during the training of the model). By using the same scaler on the input data, we ensure the input data will be in the same format as the model is trained to receive them, and we will therefore get consistent and accurate predictions.

C. Prediction Module

The Prediction Module consists of 4 different algorithms (Logistic Regression, Decision Tree, Random Forest and XGBoost) which were trained and evaluated during the development phase. The training notebook evaluated them based on their accuracy scores, and then automatically selected the best-performing model (i.e. the model with the highest accuracy score). The final selected model for the study was the XGBoost model.

XGBoost is a collection of models (trees) that build new models one at a time based on the previous model's errors; therefore, it has the highest potential for accuracy when predicting data in a structured tabular format (such as a patient's medical history). The output for each inputted record is a probability score; meaning a proportion of the probability returned will be between the range of 0-1 and each proportion corresponds to one of three (3) risk categories i.e., HIGH = equal to or greater than 0.15, MEDIUM = between 0.05 and 0.15 and LOW = less than 0.05.

D. Model Transparency Module

This module utilized SHAP for clarifying predictions that are specific to individuals by determining how factors relating to individual input variable values impact their distance from each of the predicted model probabilities in an ensemble of models. If the SHAP value for individual features is positive, then that value contributes positively to the predicted probability, and conversely. We utilized the TreeExplainer from SHAP, because it provides the exact Shapley values for tree models without the need for approximation calculations. In addition, the application utilizes a horizontal bar chart to display the contribution that each prediction of an individual user has on their predicted probability; i.e., the greatest contributing input variables are displayed on the left side of the chart, and the least contributing input variables on the right side. Finally, a second global feature importance chart will provide a global context in relation to the global training set.

E. Clinical Summary Module

Once the SHAP values have been calculated, the next thing that will happen is to find the three features that have the highest absolute values. These three features will then be used to create a short, written "note" for each of those features based on the data entered. For example, if we have HbA1c as the highest contributing feature, and the user entered a value of 7.2 for HbA1c, then the report would indicate that HbA1c is the primary variable for predicting the risk of a diabetic diagnosis and that a level of 7.2 is within the diabetic range. If the next variable, blood glucose, comes in at 180, then the report would note that this is a moderate or elevated blood glucose concentration and will contribute positively to the user's risk classification. Finally, if the user's BMI is equal to 31, it will then be flagged as an obese BMI and thus also contributepositive to the user's overall risk profile. Following this process will not oniy give the user a label (the risk) but will also allow the user to understand the three health "indicators" associated with their risk and why they are important clinically.

F. Recommendation Module

The Recommendation Module processes the same inputs and risk classification and provides two different types of output. The 1st part of the Recommendation Module contains comments for each feature in accordance with the input values. For example; a patient 45 or older is recommended to have a screening every 6 months; an individual who has a Body Mass Index (BMI) of greater than 30 is provided with a weight loss plan; and someone is currently smoking is strongly advised to quit immediately. The 2nd part of the Recommendation Module provides a dietary plan that correlates with the level of risk for the patient. High risk patients will receive an extremely regimented dietary plan which includes eating leafy green vegetables, whole grains,

fruits with a low glycemic index, boiled or grilled sources of protein, and adequate fluids; while avoiding foods high in sugar content, white rice and processed foods. Individuals that are classified as at medium risk will receive a well-balanced dietary plan that consists of providing a high fiber diet from home-cooked meals. Finally, low risk patients will simply be recommended to continue on with what they are currently doing.

G. System Workflow

When the user clicks the predict button on their application, the input values entered through the sidebar get compiled into a dataframe that matches the column structure used in the training data. The input dataframe is then passed through a scaler to prepare it for input into an XGBoost model, where probability scores and SHAP values are calculated sequentially, stored in session state and available to all individual users as they navigate through each output tab (feature importance charts, SHAP explanation, clinical summary/recommendations). The complete time elapsed from when a button is pressed until the output is displayed is less than one second.

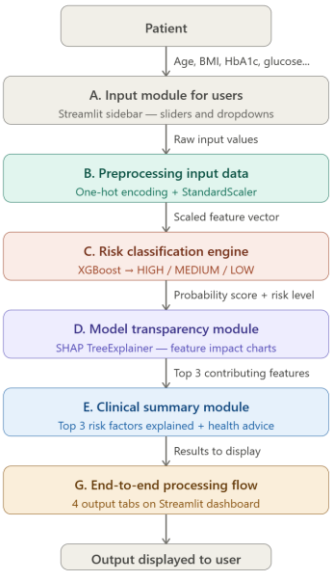

Fig 4.1 Workflow Diagram

V. SYSTEM ARCHITECTURE & IMPLEMENTATION

The Diabetes Risk Prediction System has been built with modularity and layered architecture to offer a seamless web-based solution integrating user interaction, data processing, machine-learning prediction, explainable AI, visualization and generating personalized recommendations into one platform. A structured workflow guides the processes from collection of user input via an interface at the frontend, processing by the backend, analysis via a trained machine-learning model and delivery of interpretable output and personalized recommendations to users. This architecture provides an adaptable and real-time scalable solution that meets the needs of general users and healthcare applications alike.

A. Frontend and Backend Processing: The presentation layer of the system is created using the Streamlit framework; this allows rapid development of a user friendly, interactive, and data entry interface as well as a visualization of results. The user or end user will input a number of health-related parameters, including Age, BMI, HbA1c, Blood Glucose, Hypertension and Heart Diseases, as well as Gender and Smoking Habits through the use of slider bars, drop-down user controls and selection fields from the presentation layer (user interface) of the health prediction application; result prediction (i.e., output) from a prediction model will also be displayed along with a graphical explanation of the result, and the user will receive guidelines/recommendations based on their input through the presentation layer of the user interface.

The backend of the system is designed using Python and is responsible for performing three main tasks: data preprocessing, executing the model(s) associated with the predictions that will be provided to the end user (e.g., output), and generating outputs. Preprocessed input data is then transformed based on feature alignment, encoding, and scaling so that it is consistent with the training dataset by using a StandardScaler (in conjunction with a previously trained model) that was trained on the health prediction application's training dataset. Furthermore, the backend manages communications that occur between the various modules of the health prediction application: prediction module, explainability module, visualization module, recommendation generation module, etc.

B. Machine Learning and Prediction: The system employs an XGBoost (Extreme Gradient Boosting) classifier for predicting a user's risk of developing diabetes due to its solid accuracy rate, efficacy, and proficiency in processing and analyzing structured healthcare data. The XGBoost model was trained on a dataset that possesses numerous attributes related to human health, which was saved as a joblib file on disk and is being available for real-time prediction.

The preprocessed input data for the prediction is passed directly to the XGBoost model, which evaluates the interactions between the various features of the dataset to determine a probability score, or the likelihood of diabetes. Based on the derived probability score, the system classifies users into defined categories of diabetes risk. Use of the XGBoost classifier leads to improved performance in the prediction of diabetes risk, reduced overfitting of the prediction model, and thus contributes to providing valid and accurate predictions for the early detection of diabetes.

C. Workflow and System Output:This system uses the SHAP library to provide explanations for model predictions through Explainable Artificial Intelligence by providing inputs of each feature into the prediction model. The visualizes are created using pre-defined plotting functions from Matplotlib and SHAP, and each of the features can have importance plotted and also be plotted for an individual user.

This system builds its personalized recommendations on prediction results and provides personalized recommendations for dietary, lifestyle, and medical changes to promote preventive care. The workflow starts with the

collection of user information and goes through four steps: preprocessing on the server, prediction from the XGBoost model, generating explanation and visualizations from both the SHAP and Matplotlib, and outputting the recommendations. All of these actions happen in real time so that user experience is efficient and easy-to-navigate.

VI. RESULTS

A. Experimental Setup: The system proposed to predict diabetes was tested on the sample of 100,000 patient records (80,000 training and 20,000 test samples) with stratified sampling. Training of all models was performed with the same preprocessing conditions one-hot encoding of categorical features and StandardScaler normalization of numerical features. The diabetic class was measured using Accuracy, ROC-AUC Score, Precision, Recall, and F1-Score measures of performance.

B. Model Performance Comparison: The performance of the four classifiers on the test set was summarized in the table below:

Models	**Accuracy**	**ROC-AUC**	**Precision (Diabetic)**	**Recall (Diabetic)**	**F1-Score (Diabetic)**
Logistic Regression	88.89 %	0.9630	0.4268	0.8941	0.5778
Decision Tree	88.44 %	0.9742	0.4189	0.9288	0.5774
Random Forest	92.80 %	0.9759	0.5483	0.8676	0.6720
XGBoost	**97.21 %**	**0.9796**	**0.9719**	**0.6924**	**0.8087**

The highest accuracy of 97.21, the highest ROC-AUC of 0.9796, indicated that XGBoost is the automatically chosen final model to be deployed.

C. Discussion of Results: The accuracy of the Logistic Regression was 88.89% and the high recall of the diabetic class was 0.8941, that is it has correctly identified most of the diabetic patients. Nevertheless, its accuracy of 0.4268 suggests that it has a large number of false positives - a significant number of non-diabetic patients were wrongly labeled. This is symptomatic of the model being over-confident in positives since the model has a linear decision boundary. The same trend was reflected in Decision Tree (88.44% accuracy, recall 0.9288) which is highly sensitive but lacks precision (0.4189). Although it was useful in interpretability, it performed poorly due to overfitting. Random Forest was very much better than single models with accuracy and F1-score of 92.80 and 0.6720 respectively on diabetics. Its 400 decision trees did a good job of reducing variance and better addressing the imbalance of the classes, which makes it a powerful baseline ensemble technique. XGBoost had 97.21 percent accuracy, precision of 0.9719 and F1-score of 0.8087, which was better than all other models. Its gradient boosting algorithm, together with regularization parameters (learning_rate=0.05, subsample=0.85) enabled it to capture the interaction of complex features without overfitting. The great accuracy implies that XGBoost is right 97.19 percent of the time when they predict a patient to be diabetic - which in clinical terms is crucial to reduce the number of unnecessary patients worrying and incurring follow-up expenses. The ROC-AUC scores of all models were over 0.96, which means that all models have a high discriminative ability regardless of the classification thresholds. The XGBoost has a ROC-AUC of 0.9796 which ascertains its strength as the best overall classifier.

D. Feature Importance (SHAP Analysis): SHAP analysis of the XGBoost model demonstrated that the highest impact of predictors of diabetes was the HbA1c level and blood glucose level, which are validated in clinical guidelines. BMI and age had a moderate effect with a smaller but significant effect of hypertension and heart disease. The individual predictive value of smoking history and gender was the weakest, but these factors played a part in the overall discrimination ability of the model. These results are quite consistent with medical knowledge, and the clinical validity of proposed system is proved.

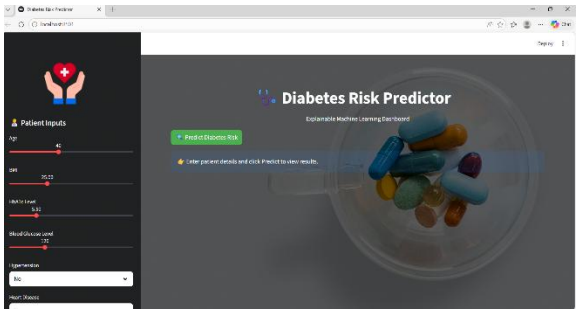

Fig 6.1 Home Page

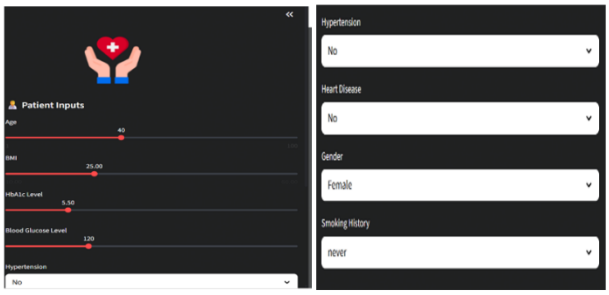

Fig 6.2User Input Screen

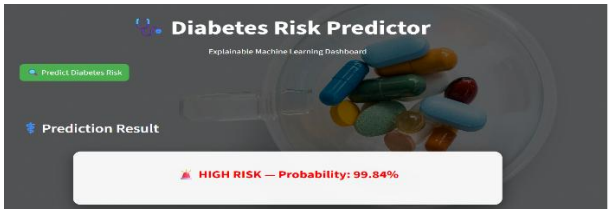

Fig 6.3 Prediction Screen

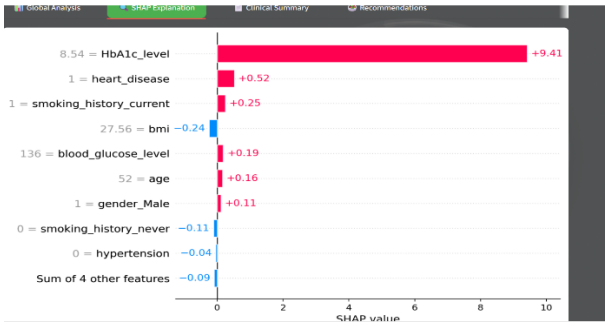

Fig 6.4 SHAP Explaination

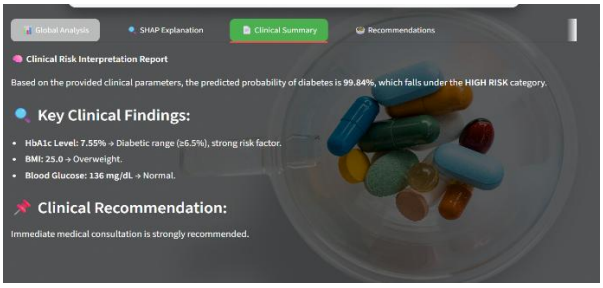

Fig 6.5 Clinical Summary

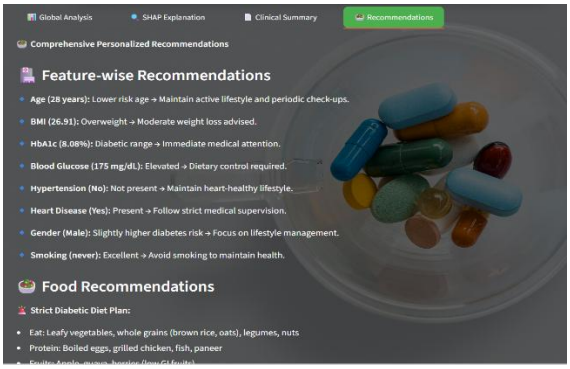

Fig 6.5 Recommendation Screen

VII. CONCLUSION

In this paper, an end-to-end machine learning-based diabetes prediction and clinical decision support system was demonstrated that was trained on 100,000 patient records. There were four classification algorithms: Logistic Regression, Decision Tree, Random Forest and XGBoost, which were trained, assessed, and compared in a systematic manner. The XGBoost model was the most effective with an accuracy of 97.21% and a ROC-AUC of 0.9796 which far exceeded the baseline models of traditional methods.

One of the contributions of this work is the combination with SHAP-based explainability, giving transparent, feature-level explanations behind the individual predictions. This is a solution to one of the most significant obstacles to clinical adoption of AI systems - the so-called black box problem as it enables health professionals to know why a patient is deemed high risk. The system produces risk stratified (Low / Medium / High) and actionable clinical reports, and is practically applicable to the current healthcare practice environment. The use of the system as an interactive Streamlit web application also reduces the barrier to adoption, allowing both clinicians and patients to use predictions without any technical skills. This system is an important contribution to AI-assisted early diabetes screening due to its combination of high predictive accuracy, model interpretability, and ease of deployment.

Overall, the suggested system shows that machine learning, in particular, gradient boosting algorithms, such as XGBoost, and explainability models and user-friendly implementation can become a powerful, dependable, and comprehensible instrument to prevent and detect diabetes early on.

VIII. FUTURE SCOPE

The existing system offers a solid base to build AI-based diabetes prediction, and a number of directions can further enhance its features:

a. Deep Learning Implementation: Future research could examine deep neural networks like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) on time-varying patient data, which may be able to capture longitudinal health patterns that cannot be modeled by a traditional ML model.
b. Prediction of Multiple Diseases: Extending the model of multiple diseases to police other chronic diseases occurring with diabetes (renal failure, cardiovascular risks & diabetic retinopathy) and develop an Integrated Metabolic Risk Assessment.
c. Real-Time EHR Interface: The system could be smoothly connected to the Electronic Health Records (EHR) systems like Epic or FHIR-compliant databases to enable real-time, automated risk screening at the point of care, without a need to enter the data manually.
d. Federated Learning to protect privacy: Federated learning can be adopted to train models in more than one hospital without collecting the privacies of patients to achieve larger and more diverse training data sets without compromising the privacy of patients.
e. Wearable Data: Adding real-time continuous data streams of wearable devices (smartwatches, CGMs Continuous Glucose Monitors) would allow dynamically tracking the risk of diabetes and providing an early warning to patients.
f. Greater and More Diverse Datasets: The models utilized in the future need to be tried on larger datasets with a greater degree of geographical and ethnical heterogeneity to reduce the chances of biasness and the generalization of the results to other population groups.
g. Mobile Application Deployment: Mobileizing the Streamlit web application (Android/iOS) would be a major step to making the tool more accessible, particularly to primary care doctors and patients in low-resource environments.
h. Causal Inference Methods: Future work can use causal inference methods in addition to predictive modeling to determine risk factors that can be modified, allowing more targeted preventive interventions than the simple forecasting of risk.

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